

Powers, Roots and Standard Form



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Index laws for numbers, then standard form:
 $A \times 10^n$ with $1 \leq A < 10$. Convert, calculate, and
 compare big and small numbers.

THE TOPIC ON ONE MAP

Powers, Roots and Standard Form

01 Powers and roots of numbers

$$5^0, 3^{-2}$$

$$125^{\frac{2}{3}}, 27^{-\frac{2}{3}}$$

$$\sqrt{2} \div \frac{8^3}{16^{3/2}} = 2^n$$

02 Standard form: convert

76 000 000

$$5.4 \times 10^{-4}$$

which is in standard form?

order them

03 Standard form: calculate

$$(5 \times 10^{42}) \times (7 \times 10^{-180})$$

$$\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$$

$$4\% \text{ of } 4.5 \times 10^{157}$$

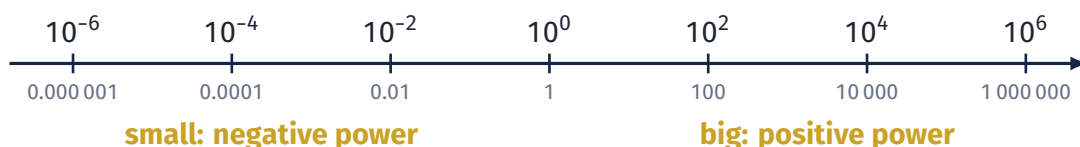
04 Tables and real data

difference of two oceans

"k times as big"

per person

THE SLIDING SCALE The power of 10 counts how far the decimal point moves:



01 Powers and Roots of Numbers

Three rules turn any strange power into a normal number: zero $\rightarrow$ 1, negative $\rightarrow$ flip, fraction $\rightarrow$ root then power.

WHEN YOU SEE $\rightarrow$ YOU DO

a^0 (anything to the power 0)

$\rightarrow = 1$

3^{-2} (negative power)

$\rightarrow$ flip: $\frac{1}{3^2} = \frac{1}{9}$

$125^{\frac{2}{3}}$ (fraction power)

$\rightarrow$ bottom = root, top = power: $(\sqrt[3]{125})^2 = 25$

“write as a power of 2 / find n ”

$\rightarrow$ write every number as a power of 2, then use the laws

“show that”

$\rightarrow$ show the root step and the power step

Negative? Flip

Root (bottom)

Power (top)

WORKED MODEL • NEGATIVE FRACTION POWER

Work out $27^{-\frac{2}{3}}$.

1 $\left(\frac{1}{27}\right)^{\frac{2}{3}}$

flip

2 $\left(\frac{1}{3}\right)^2$

cube root

ANSWER $\frac{1}{9}$

× **TRAP** $64^{\frac{1}{4}}$ is **not** $\frac{1}{4}$ of 64. It is the fourth root: $\sqrt[4]{64} = 2\sqrt{2}$.

IDEA 01

Practise: Powers and Roots of Numbers

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Three rules

CH 1 • TOPIC 4 3 marks

Work out (a) 2^{-3} (b) $27^{\frac{2}{3}}$ (c) $16^{-\frac{1}{2}}$

SELECTED PRACTICE

Q2 PRACTICE Practice • Easy

CH 1 • TOPIC 4 2 marks

1 Write these numbers in order of size. Start with the smallest number.

5^{-1}	0.5	-5	5^0
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(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 4 1 mark

14 Circle the expression that has the **largest** value when $a < -1$

$\frac{1}{2}a$	a	a^2	a^3
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(1 mark)

Q4**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 2 marks

1 (a) Show that $2^{-3} = \frac{1}{8}$

(1 mark)

(b) $5\sqrt{5}$ can be written in the form 5^k .

Find the value of k .

(1 mark)

Q5**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4

3 marks

14 Show that $\frac{\sqrt[3]{81}}{3}$ can be written as $3^{\frac{1}{3}}$

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q15

CH 1 • TOPIC 4

6 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

 $n = \dots\dots\dots$

(b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

 $\dots\dots\dots$ **(Total for Question 15 is 6 marks)**

02 Standard Form: Convert

Standard form is $A \times 10^n$ with $1 \leq A < 10$. Move the point until one non-zero digit is in front of it; count the moves.

WHEN YOU SEE → YOU DO

“write in standard form”, a big number

→ power = number of places moved left (positive)

0.000068 (small number)

→ negative power: 6.8×10^{-5}

“as an ordinary number”

→ positive power → bigger; negative power → smaller

“circle the number in standard form”

→ check $1 \leq A < 10$: 0.9×10^{-3} and 12×10^7 are not

“write in order of size”

→ convert all to ordinary numbers (or the same power), then order

ZONE BIG

$$76\,000\,000 = 7.6 \times 10^7$$

the point moved 7 places left

ZONE SMALL

$$0.000\,54 = 5.4 \times 10^{-4}$$

the point moved 4 places right

× **TRAP** 0.38×10^{-1} and 3800×10^{-4} are not in standard form — convert before you compare.

IDEA 02

Practise: Standard Form: Convert

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Both directions

CH 1 • TOPIC 4

2 marks

(a) Write 0.003 05 in standard form. (b) Write 4.6×10^5 as an ordinary number.

SELECTED PRACTICE**Q8****PRACTICE**

Practice • Easy

CH 1 • TOPIC 4

2 marks

4 (a) Write 7.8×10^{-4} as an ordinary number.

(1 mark)

(b) Write 95 600 000 as a number in standard form.

(1 mark)

Q9**PRACTICE**

Practice • Easy

CH 1 • TOPIC 4 2 marks

8 (a) Write 0.00562 in standard form.

(1 mark)

(b) Write 1.452×10^3 as an ordinary number.

(1 mark)

Q10**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

3 (a) Write down the value of 10^0 .

(1 mark)

(b) Write down the value of 10^{-2} .

(1 mark)

(c) Write these numbers in order of size.
Start with the smallest number.

2.73×10^3	27.3×10^{-3}	273×10^2	0.00273
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(2 marks)

Q11**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 4 marks

10 (a) Write 8.2×10^5 as an ordinary number.

(1 mark)

(b) Write 0.000 376 in standard form.

(1 mark)

(c) Work out the value of $(2.3 \times 10^{12}) \div (4.6 \times 10^3)$
Give your answer in standard form.

(2 marks)

PAST PAPERS

Q12**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q5

CH 1 • TOPIC 4 4 marks

5 (a) Write 5.7×10^{-3} as an ordinary number.

.....

(b) Write 800 000 in standard form.

.....

(c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....

(Total for Question 5 is 4 marks)**Q13****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q6

CH 1 • TOPIC 4 2 marks

6 (a) Write 2 840 000 000 in standard form.

.....

(b) Write 2.5×10^{-4} as an ordinary number.

.....

(Total for Question 6 is 2 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q8

CH 1 • TOPIC 4 4 marks

8 (a) Write 5×10^4 as an ordinary number.

.....

(b) Write 0.00006 in standard form.

.....

(c) Work out $(4 \times 10^{512}) \div (1.6 \times 10^{700})$

Give your answer in standard form.

.....

(Total for Question 8 is 4 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q8

CH 1 • TOPIC 4 2 marks

8 (a) Write 0.000089 in standard form.

.....

(b) Write 8.34×10^4 as an ordinary number.

.....

(Total for Question 8 is 2 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q6

CH 1 • TOPIC 4 2 marks

6 (a) Write 76 000 000 in standard form.

.....

(b) Write 5.4×10^{-4} as an ordinary number.

.....

(Total for Question 6 is 2 marks)**Q17****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q9

CH 1 • TOPIC 4 4 marks

9 (a) Write 5.87×10^{-4} as an ordinary number.

.....

(b) Write 84 000 000 in standard form.

.....

The number of neurons in a human brain is 8.5×10^{10}

The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

(c) Work out the value of K

Give your answer correct to one decimal place.

 $K =$ **(Total for Question 9 is 4 marks)**

Q18**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q9

CH 1 • TOPIC 4 4 marks

9 (a) Write 6.04×10^5 as an ordinary number.

.....

(b) Write 0.000 07 in standard form.

.....

(1)

(c) Work out $\frac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

.....

(Total for Question 9 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q8

CH 1 • TOPIC 4 4 marks

8 (a) Write 520 million in standard form.

.....

(b) Write 8.79×10^{-5} as an ordinary number.

.....

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....

(Total for Question 8 is 4 marks)

03 Standard Form: Calculate

Multiply or divide: numbers with numbers, powers with powers — then fix A. Add or subtract: use the calculator (or make the powers the same).

WHEN YOU SEE → YOU DO

$$(5 \times 10^{42}) \times (7 \times 10^{-180})$$

$$\longrightarrow 35 \times 10^{-138} \rightarrow \text{fix to } 3.5 \times 10^{-137}$$

÷ **in standard form**

→ divide the numbers, subtract the powers

“+” or “-”

→ calculator ($\times 10^x$ key), or same power first

$$(6 \times 10^{40})^3$$

$$\longrightarrow 6^3 \times 10^{120} = 216 \times 10^{120} \rightarrow 2.16 \times 10^{122}$$

“give your answer in standard form”

→ last line must have $1 \leq A < 10$

Numbers together

Powers together

Fix A

Check $1 \leq A < 10$

WORKED MODEL • JUNE 2025 P1H

Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$.

1 $(5 \times 7) \times 10^{42-180}$

numbers · powers

2 35×10^{-138}

not yet standard form

ANSWER 3.5×10^{-137}

CALCULATOR CHECK Use the $\times 10^x$ key, never “ $\times 10^{\wedge}$ ”. Put a fraction’s top and bottom in brackets.

IDEA 03

Practise: Standard Form: Calculate

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM Fix A

CH 1 • TOPIC 4 2 marks

Work out $(3 \times 10^4) \times (4 \times 10^5)$. Give your answer in standard form.

Q21 DR ESLAM Divide

CH 1 • TOPIC 4 2 marks

Work out $(6 \times 10^8) \div (2 \times 10^{-3})$ in standard form.

SELECTED PRACTICE**Q22** PRACTICE Practice • Easy

CH 1 • TOPIC 4 2 marks

- 5 Work out the value of $(7.5 \times 10^4) \times (2.5 \times 10^3)$
Give your answer in standard form.
-

(2 marks)

Q23**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 3 marks

- 1 One sheet of paper is 9×10^{-3} cm thick.

Mark wants to put 500 sheets of paper into the paper tray of his printer.
The paper tray is 4 cm deep.

Is the paper tray deep enough for 500 sheets of paper?
You must explain your answer.

(3 marks)

Q24**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 9 marks

- 16 (a)** A company makes sweets.
The sweets are put into packets.

Here are some facts.

1.47×10^7	3.5×10^5
sweets are made every day	packets of sweets are produced every day

Calculate the mean number of sweets in one packet.

(2 marks)

- (b)** Sweets are made on 288 days each year.

Calculate the number of sweets made each year.
Give your answer in standard form.

(3 marks)

- (c)** The company has 152 machines making the sweets. Each machine operates for 15 hours each day.

- i) Calculate the number of sweets made by one machine each hour.
Give your answer as an ordinary number correct to the nearest 10.

[3]

- ii) State one assumption you have made in part **(c)(i)**.

[1]

(4 marks)

Q25**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 2 marks

- 11 Work out the value of $(3.5 \times 10^6) \div (5 \times 10^{-3})$.
Give your answer in standard form.

(2 marks)**PAST PAPERS****Q26****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q6

CH 1 • TOPIC 4 3 marks

- 6 (a) Write 7.8×10^{-4} as an ordinary number.

(b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

(Total for Question 6 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q6

CH 1 • TOPIC 4 4 marks

- 6 (a) Write down the value of y^0

.....
(1)

- (b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

.....
(Total for Question 6 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q8

CH 1 • TOPIC 4 2 marks

8

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

.....
(Total for Question 8 is 2 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q12

CH 1 • TOPIC 4 3 marks

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

.....

(Total for Question 12 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q8

CH 1 • TOPIC 4 3 marks**8** (a) Write 5.6×10^{-3} as an ordinary number.

(b) Work out
$$\frac{6 \times 10^3}{2.1 \times 10^{-4} + 9 \times 10^{-5}}$$

Give your answer in standard form.

.....

(Total for Question 8 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q9

CH 1 • TOPIC 4 3 marks

9 (a) Write 8.4×10^{-5} as an ordinary number.

.....

(b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....

(Total for Question 9 is 3 marks)**Q32****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q14

CH 1 • TOPIC 4 2 marks

14 Work out $6.7 \times 10^{135} + 3 \times 10^{134}$
Give your answer in standard form.

.....

(Total for Question 14 is 2 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q8

CH 1 • TOPIC 4 3 marks

8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(Total for Question 8 is 3 marks)

04 Tables and Real Data

The numbers are in a table: read the right row, then do exactly the operation the words ask for.

WHEN YOU SEE → YOU DO

“the difference between”

→ bigger – smaller

“ k times the surface area”, “how many times further”

→ divide

“per person”, “bacteria per raindrop”

→ divide by the number of people / drops

“correct to the nearest whole number / 2 s.f.”

→ round only at the end

WORKED MODEL • JANUARY 2020 P2HR

Pacific $1.56 \times 10^8 \text{ km}^2$, Arctic $1.41 \times 10^7 \text{ km}^2$. Pacific = $k \times$ Arctic. Find k to the nearest whole number.

$$1 \quad k = \frac{1.56 \times 10^8}{1.41 \times 10^7}$$

“times” → divide

$$2 \quad k = 11.06 \dots$$

calculator

ANSWER $k = 11$

IDEA 04

Practise: Tables and Real Data

My questions first, then selected practice, then real past papers.

Q34 **DR ESLAM** How many times

CH 1 • TOPIC 4 **2 marks**

The Earth's mass is 5.97×10^{24} kg; the Moon's is 7.35×10^{22} kg. How many times heavier is the Earth? (1 d.p.)

SELECTED PRACTICE

Q35**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

6 (a) The table shows some information about eight planets.

Planet	Distance from Earth (km)	Mass (kg)
Earth	0	5.97×10^{24}
Jupiter	6.29×10^8	1.898×10^{27}
Mars	7.83×10^7	6.42×10^{23}
Mercury	9.17×10^7	3.302×10^{23}
Neptune	4.35×10^9	1.024×10^{26}
Saturn	1.28×10^9	5.68×10^{26}
Uranus	2.72×10^9	8.683×10^{25}
Venus	4.14×10^7	4.869×10^{24}

Write down the name of the planet with the greatest mass.

(1 mark)

(b) Find the difference between the mass of Venus and the mass of Mercury.

(1 mark)

(c) Nishat says that Neptune is over a hundred times further away from Earth than Venus is.

Is Nishat right?

You must show how you get your answer.

(2 marks)

Q36**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

- 17 (a)** The table shows the population, correct to two significant figures, of each of six countries in April 2016.

Country	Population (April 2016)
Hungary	9.8×10^6
Mexico	1.3×10^8
Thailand	6.8×10^7
Nigeria	1.9×10^8
Singapore	5.7×10^6
Egypt	9.3×10^7

Write 9.3×10^7 as an ordinary number.

(1 mark)

- (b)** Which of these countries had the least population?

(1 mark)

- (c)** The population of China was 1.382×10^9 in April 2016.
The population of India was 1.327×10^9 in April 2016.

Work out the difference between the population of China and the population of India in April 2016.

Give your answer in standard form.

(2 marks)

PAST PAPERS

Q37 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2HR • Q10

CH 1 • TOPIC 4 3 marks

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

- (a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean.
Give your answer in standard form.

..... square kilometres

The surface area of the Pacific Ocean is k times the surface area of the Arctic Ocean.

- (b) Work out the value of k .
Give your answer correct to the nearest whole number.

$k =$

(Total for Question 10 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q8

CH 1 • TOPIC 4 3 marks

- 8 The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

- (a) Which ocean has the greatest length of coastline?

.....

- (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline.
Give your answer in standard form.

..... km

(Total for Question 8 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q8

CH 1 • TOPIC 4 4 marks

- 8 The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

.....

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

- (b) work out the value of k .
Give your answer correct to the nearest whole number.

 $k =$

(Total for Question 8 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q9

CH 1 • TOPIC 4 5 marks

- 9 A rainwater tank contains 2.4×10^7 raindrops.

The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules

(Total for Question 9 is 5 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q10

CH 1 • TOPIC 4 3 marks

- 10** The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

Country	Total population	Total spent on healthcare (\$)
Austria	8.7×10^6	4.2×10^{10}
Luxembourg	6.3×10^5	3.7×10^9

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.

Give your answer correct to the nearest whole number.

..... dollars

(Total for Question 10 is 3 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q9

CH 1 • TOPIC 4 4 marks

- 9 The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

- (a) Write 8.8×10^6 as an ordinary number.
.....
- (b) Which of these cities had the least population in 2018?
.....
- (c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018
Give your answer in standard form correct to 2 significant figures.
.....

(Total for Question 9 is 4 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q9

CH 1 • TOPIC 4 3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes

(Total for Question 9 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 0

$$a^0 = 1$$

always

ZONE -

$$a^{-n} = \frac{1}{a^n}$$

flip

ZONE $\frac{m}{n}$

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

root, then power

- 01 Powers of numbers** Zero $\rightarrow$ 1, negative $\rightarrow$ flip, fraction $\rightarrow$ root then power.
- 02 Convert** $A \times 10^n$, $1 \leq A < 10$; big $\rightarrow +n$, small $\rightarrow -n$.
- 03 Calculate** Numbers with numbers, powers with powers, then fix A.
- 04 Tables** Difference = subtract; “times” / “per” = divide; round last.

BOTTOM LINE Standard form always ends with $1 \leq A < 10 \rightarrow$ check it on every last line.

Answers

Powers, Roots and Standard Form • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • POWERS AND ROOTS OF NUMBERS

- Q1** Dr Eslam • Three rules (a) $\frac{1}{8}$ (b) 9 (c) $\frac{1}{4}$
- Q2** Practice • Easy $-5, 5^{-1}, 0.5, 5^0$
- Q3** Practice • Easy a^2
- Q4** Practice • Hard (a) $\frac{1}{2^3}$ (b) $k = \frac{3}{2}$
- Q5** Practice • Hard $\frac{3^{\frac{4}{3}}}{3} = 3^{\frac{1}{3}}$
- Q6** Edexcel 4MA1 • May 2023 • Paper 1H • Q15 (a) $n = -\frac{5}{2}$ (b) 1.8×10^{156}

IDEA 02 • STANDARD FORM: CONVERT

- Q7** Dr Eslam • Both directions (a) 3.05×10^{-3} (b) 460 000
- Q8** Practice • Easy (a) 0.000 78 (b) 9.56×10^7
- Q9** Practice • Easy (a) 5.62×10^{-3} (b) 1452
- Q10** Practice • Medium (a) 1 (b) 0.01 (c) 0.00273, 27.3×10^{-3} , 2.73×10^3 , 273×10^2
- Q11** Practice • Hard (a) 820 000 (b) 3.76×10^{-4} (c) 5×10^8
- Q12** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q5 (a) 0.0057 (b) 8×10^5 (c) 4.55×10^6
- Q13** Edexcel 4MA1 • January 2021 • Paper 1H • Q6 (a) 2.84×10^9 (b) 0.000 25
- Q14** Edexcel 4MA1 • January 2022 • Paper 1HR • Q8 (a) 50 000 (b) 6×10^{-5} (c) 2.5×10^{12}
- Q15** Edexcel 4MA1 • June 2022 • Paper 1H • Q8 (a) 8.9×10^{-5} (b) 83 400
- Q16** Edexcel 4MA1 • June 2023 • Paper 2H • Q6 (a) 7.6×10^7 (b) 0.000 54
- Q17** Edexcel 4MA1 • November 2023 • Paper 1H • Q9 (a) 0.000 587 (b) 8.4×10^7 (c) $K = 57.8$
- Q18** Edexcel 4MA1 • June 2024 • Paper 2HR • Q9 (a) 604 000 (b) 7×10^{-5} (c) 2×10^5
- Q19** Edexcel 4MA1 • June 2025 • Paper 1H • Q8 (a) 5.2×10^8 (b) 0.000 087 9 (c) 3.5×10^{-137}

IDEA 03 • STANDARD FORM: CALCULATE

- Q20** Dr Eslam • Fix A $12 \times 10^9 = 1.2 \times 10^{10}$
- Q21** Dr Eslam • Divide 3×10^{11}
- Q22** Practice • Easy 1.875×10^8
- Q23** Practice • Medium $500 \times 0.009 = 4.5 \text{ cm} > 4 \text{ cm}$: No
- Q24** Practice • Medium (a) 42 (b) 4.2336×10^9 (c)(i) 6450 (ii) every machine works at the same rate all day
- Q25** Practice • Hard 7×10^8
- Q26** Edexcel 4MA1 • January 2020 • Paper 1H • Q6 (a) 0.000 78 (b) 2.25×10^7
- Q27** Edexcel 4MA1 • January 2021 • Paper 2H • Q6 (a) 1 (b) 3.2×10^{125}
- Q28** Edexcel 4MA1 • January 2022 • Paper 1H • Q8 1.26×10^{122}
- Q29** Edexcel 4MA1 • January 2022 • Paper 2H • Q12 2.16×10^{122}
- Q30** Edexcel 4MA1 • May 2023 • Paper 1HR • Q8 (a) 0.0056 (b) 2×10^9
- Q31** Edexcel 4MA1 • November 2024 • Paper 1H • Q9 (a) 0.000 084 (b) 5.2×10^{146}
- Q32** Edexcel 4MA1 • June 2025 • Paper 1HR • Q14 7×10^{135}
- Q33** Edexcel 4MA1 • November 2025 • Paper 2H • Q8 (a) 57 600 (b) 7.7×10^6

IDEA 04 • TABLES AND REAL DATA

- Q34** Dr Eslam • How many times 81.2
- Q35** Practice • Medium (a) Jupiter (b) 4.5388×10^{24} kg (c) $4.35 \times 10^9 \div 4.14 \times 10^7 = 105$: yes
- Q36** Practice • Medium (a) 93 000 000 (b) Singapore (c) 5.5×10^7
- Q37** Edexcel 4MA1 • January 2020 • Paper 2HR • Q10 (a) 3.74×10^7 (b) $k = 11$
- Q38** Edexcel 4MA1 • January 2021 • Paper 2HR • Q8 (a) Pacific (b) 9.393×10^4
- Q39** Edexcel 4MA1 • May 2021 • Paper 1H • Q8 (a) 1.318×10^9 (b) $k = 11$
- Q40** Edexcel 4MA1 • November 2021 • Paper 2H • Q9 (a) 0.073 (b) 4×10^{28}
- Q41** Edexcel 4MA1 • January 2022 • Paper 2HR • Q10 \$1045
- Q42** Edexcel 4MA1 • June 2022 • Paper 2HR • Q9 (a) 8 800 000 (b) Barcelona (c) 2.9×10^7
- Q43** Edexcel 4MA1 • May 2024 • Paper 1H • Q9 (a) 35 000 000 (b) 7.6×10^6 tonnes